



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

Imię i nazwisko / Name, surname	Grzegorz Piotrowski
Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
Nr ORCID	0009-0004-6013-0461
Promotor/ promotorzy / Supervisor(s)	dr hab. inż. Tomasz Jan Kajdanowicz, prof. uczelni, Politechnika Wrocławska (K46, W4), ORCID 0000-0002-8417-1012
Promotor pomocniczy / Assistant supervisor	—

2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

PL: Uczenie modeli nawigacji robotów w cyfrowych bliźniakach przy mniejszej ilości danych rzeczywistych: metoda typu rzeczywistość–symulacja–rzeczywistość

EN: Learning robot navigation models in digital twins with less real data: a real-to-sim-to-real method

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	<i>(Oct 2025 – Feb 2026)</i> Research target: define the problem and map the state of the art. T1.1 Systematic literature review in four areas: (a) sim-to-real transfer and domain randomization, (b) learning under distribution shift and domain adaptation, (c) neural scene reconstruction (NeRF, 3D Gaussian Splatting), (d) learning-based visual navigation. <i>D:</i> annotated bibliography and a written research-gap statement (basis of §6). T1.2 Identification of the open problem: how much real data sim-to-real transfer needs, and whether reconstructed scenes can replace real data. <i>D:</i> problem statement and preliminary research questions, discussed with the supervisor. T1.3 Doctoral School courses of semester 1. <i>D:</i> completed courses. Place: K46.



Semestr 2	(Mar – Sep 2026) Research target: formulate the research plan and select the experimental basis. T2.1 Formulation of the data-efficient real-to-sim-to-real thesis, research questions RQ1–RQ4 (capture less, learn robustly in an imperfect twin, few real data, budget of the whole loop) and hypotheses H1–H4 with measurable decision rules and a proxy-reality evaluation protocol. <i>D</i>: §7 and §9 of this IPB. T2.2 Comparative survey of public benchmarks and datasets (e.g. ScanNet++, HM3D), simulators (e.g. Habitat-Sim) and 3D Gaussian Splatting implementations against the needs of RQ1–RQ4 (real video + reference scan, licence, compute cost). <i>D</i>: tool-selection note with a shortlist for Stage I. T2.3 Individual Research Plan agreed with the supervisor. <i>D</i>: this IPB, signed and submitted by 30.09.2026. Place: K46.
Semestr 3	(Oct 2026 – Feb 2027) Stage 1 – transfer of real data into a digital twin (RQ1, H1). T3.1 Proxy-reality set-up (§9): ≥ 20 public indoor scenes (10 held out) with laser-scan references and separate phone captures; twins at ≥ 4 capture budgets with an existing neural reconstruction pipeline and the physical properties needed for navigation; navigation tasks, learner and pretrained models running on the twins; pilot fixing task difficulty (so that P stays below its ceiling), the operator-time cost model and the GPU cost; WCSS/PLGrid computing grant; pre-registration of the protocol (§9). <i>D</i>: working pipeline and protocol in the repository, by Jan 2027. Place: K46, WCSS. T3.2 Validation stand: written agreement on robot access (planned cooperation with the K29 Denali laboratory) and sensor decision (SzD Minigrant application if a sensor must be bought). <i>D</i>: agreement, by Nov 2026. T3.3 Stage 1 method: capture guided by the navigation task and the twin's uncertainty vs. uniform, task-blind and risk-weighted selection, judged by the success of the navigation model trained in each twin. <i>D</i>: first budget–success curves (first H1 result), by Feb 2027.
Semestr 4	(Mar – Sep 2027) Stage 1 completed; Stage 2 – training navigation models in the twin (RQ1–RQ2, H1–H2). T4.1 Stage 2 method: training that follows the twin's uncertainty, with the twin extended by a learned world model and the model initialized from pretrained foundation models, vs. uniform domain randomization, at equal capture budgets. <i>D</i>: H1 completed and first H2 result (uncertainty-following training), by Apr 2027; H2 completed with the world-model extension, by Jul 2027. Place: K46, WCSS. T4.2 Prepare and submit article P1 to NeurIPS 2027 (stage 1, H1, and first stage 2 results, H2). <i>D</i>: submission, May 2027 (§11). T4.3 Stage 3 set-up: selection of real trials by the predicted twin-to-reality gap and correction code on the proxy reality (Jul–Sep 2027). <i>D</i>: first H3 result, by Sep 2027. T4.4 Grants and international activity: NCN PRELUDIUM application; poster at an international summer school; foreign co-author sought for P2; NAWA Bekker application for the sem. 6 visit. <i>D</i>: applications, poster.



Semestr 5	(Oct 2027 – Feb 2028) Mid-term evaluation (Nov 2027): autoreferat (≤ 5 pages, ~mid-Oct 2027) and 15-min presentation. Stage 3 – transfer back to reality with few real data (RQ3, H3). T5.1 Stage 3 method: active selection of real trials vs. random and twin-only failure-driven selection; correction of the twin (appearance, geometry, physical properties) and of the navigation model with the selected trials. <i>D:</i> H3 completed on the proxy reality, by Jan 2028. Place: K46, WCSS. T5.2 First robot validation campaign (2 PWr rooms, ~8 robot-hours): own captures, twins, uniform capture vs. the method. <i>D:</i> validation report. Place: Denali. Without robot access, the campaign moves to sem. 7; H1–H3 are decided on the proxy reality either way. T5.3 Article P2 (stage 3: few real data and twin correction, RQ3) submitted to ICML 2028 (~late Jan 2028; CVPR 2028, ~Nov 2027, only if H3 is complete by then). If P1 was rejected, its revision goes to CVPR 2028. <i>D:</i> submission(s).
Semestr 6	(Mar – Sep 2028) The whole pipeline (RQ4, H4, the thesis). T6.1 Real-data budget curves of the whole method vs. the strongest existing real-to-sim-to-real approach and vs. learning from real data only ("exchange rate"), with the method's settings fixed in advance. <i>D:</i> first result by Apr 2028; H4 completed, by Sep 2028. Place: K46, WCSS. T6.2 Foreign research visit (3 months, NAWA Bekker; if not funded, a 5–30-day Erasmus+ short-term mobility to a European lab; hosts in §12): joint experiments. <i>D:</i> visit report. T6.3 Article P3 (the whole pipeline against existing approaches) submitted to NeurIPS 2028 (~May 2028); if not ready, to ICLR 2029 (~late Sep 2028). <i>D:</i> submission.
Semestr 7	(Oct 2028 – Feb 2029) Consolidation (RQ1–RQ4). T7.1 Second robot validation campaign of the whole method; final analysis of H1–H4. <i>D:</i> final results. Place: K46, Denali. T7.2 Release of the method's code and experiment scripts (where dataset and model licences permit). <i>D:</i> public repository. T7.3 Optional article P4 (consolidated study) to ICLR 2029 (~late Sep 2028) or CVPR 2029 (~Nov 2028); start of dissertation writing. <i>D:</i> submission (optional), dissertation outline.
Semestr 8	(Mar – Sep 2029) T8.1 Editing of the doctoral dissertation; final version. <i>D:</i> dissertation submitted by 30.09.2029 (§4). T8.2 Responses to reviews and camera-ready versions of pending articles. <i>D:</i> revised manuscripts.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

30.09.2029



(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 / The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Mobile robots that move around buildings, such as service, delivery, inspection and assistive robots, rely increasingly on learned **navigation models**. Such models need a lot of experience from the place where they will work, and collecting it in reality is slow, expensive and sometimes unsafe. Training in simulation avoids this cost but opens a **sim-to-real gap**: a model that works in a generic simulator often fails in a real building.

Digital twins built from real data narrow this gap. Neural scene reconstruction turns a short video or depth capture of a real building into a photorealistic model of its geometry and appearance, and a few measurements give the physical properties that matter for navigation. Learned **world models** can generate further situations, and pretrained **foundation models** provide a strong starting point. This gives a **real-to-sim-to-real pipeline**: acquire real data, build a twin, train the navigation model in it at scale, and transfer it back to reality, where a little more real data checks and corrects the twin and the model. Each of these parts is an active research area. What remains open is the **real data** the whole pipeline consumes: the capture that builds the twin, the real samples that make training robust to what the twin got wrong, and the real trials that correct the result. Today these amounts are chosen by habit: data is captured uniformly, training uses generic randomization, and real trials are picked at random or by hand. How much real data a twin actually saves in navigation has, to our knowledge, not been measured systematically.

Goal of the dissertation. The goal is to develop a **method** for real-to-sim-to-real learning of navigation models that reaches a given real-world navigation performance with **significantly less real data** than existing approaches. The method covers the three stages of the pipeline: (1) deciding which real data is worth acquiring to build the twin, (2) training navigation models in the imperfect twin, extended with world models and foundation models, so that its errors do not transfer, and (3) choosing the few real data that best correct the twin and the model. The thesis is that the whole method needs at least two times less real data than the strongest existing real-to-sim-to-real approach.

Why this topic, and why in this discipline. The object is a data-efficient learning method (active learning, learning under distribution shift, uncertainty estimation, adaptation of pretrained models), not a particular robot, simulator or benchmark, which places it in *information and communication technology*. The student builds on existing open reconstruction pipelines, simulators, pretrained models and public datasets. Public scans of real buildings make it possible to count real data exactly without an own robot; trials on a real robot (planned cooperation with a PWr robotics laboratory) validate, but do not decide, the conclusions. The topic fits the Department of Artificial Intelligence (K46), whose research groups include representation learning, and the training runs on the GPU infrastructure available to PWr researchers (WCSS, PLGrid).

Potential application areas: faster and cheaper deployment of mobile robots in new buildings, such as warehouses, hospitals, offices and inspection sites, with less time spent collecting data and



supervising real trials; guidance on how much real data to collect for a twin; and, beyond robotics, any model trained on reconstructed or simulated data and used on real sensor data.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / Outline of the current state of research in the topic of the doctoral dissertation

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Simulation and sim-to-real transfer in navigation. Navigation models for mobile robots are commonly trained in simulators such as Habitat [1], where large-scale reinforcement learning essentially solves point-goal navigation after billions of steps [2]. Transfer to the real world is the open part: simulation success does not always predict real success, which motivated the Sim-vs-Real Correlation Coefficient [3], and lower-fidelity simulation can even transfer better [4]. Domain randomization varies appearance [5] or dynamics [6] uniformly within hand-set ranges, and real rollouts can adapt the simulation parameter distribution [7]. Real data still matters: in navigation, more data from a known location saturates quickly [8], and co-training on simulated and real data raises real success [9].

From real data to digital twins. Neural scene reconstruction, above all 3D Gaussian Splatting (3DGS) [10], turns a short real capture into a photorealistic model of a scene. Twins built this way train navigation models that transfer: EmbodiedSplat [11] reconstructs rooms from a phone capture and fine-tunes image-goal navigation in Habitat, Vid2Sim [12] builds interactive simulators for urban navigation from monocular video, and GaussGym [13] trains navigation and locomotion from pixels in reconstructed scenes. RialTo [14] shows the same loop for manipulation, building twins from small amounts of real data. The physical part of a twin can be identified from real trajectories as a posterior [15]. Datasets such as ScanNet++ [16] pair laser scans and DSLR images with a separate phone capture of the same scene. Building a twin is no longer the bottleneck; in these works the amount and choice of the real data that builds and corrects it are fixed by hand.

Capturing the right real data. Active view selection chooses the images that improve a reconstruction most: FisherRF [17] maximizes expected information gain, and Bayes' Rays [18] estimates an uncertainty field for a trained radiance field. Risk-aware view acquisition [19] weights FisherRF by safety-critical regions for navigation, and AREA3D [20] adds vision-language guidance. Their objective is reconstruction quality or safe exploration, not the success of a navigation model trained in the twin; we found at most two task-aware capture papers a year in 2023–2026.

Foundation models and world models. Pretrained navigation foundation models generalize across robots and environments (GNM [21], ViNT [22]), and vision-language-action models map images and instructions to actions, also for navigation [23, 24]; they still need data from the target place to work there. Learned world models predict future observations and can serve as simulators: navigation world models plan by imagining trajectories [25], and open world foundation models are post-trained per setup [26]. Recent work fine-tunes a pretrained policy in a twin reconstructed from a phone capture and uses the twin to target real rollouts [27], or improves a world model with real rollouts [28]; each reports results at a fixed, hand-chosen amount of real data. Frozen visual encoders [29] make twin and real observations comparable, and prediction-powered inference [30] estimates real performance from many cheap predictions and a few real labels.

Research gap. Real-to-sim-to-real learning of navigation models works, but the real data it consumes is set by hand and spent uniformly at every stage. Missing is one method that decides, for navigation: (i) which real data to acquire to build the twin (RQ1); (ii) how to train in an imperfect twin, extended



with world models and foundation models, so that its errors do not transfer (RQ2); (iii) which few real data to collect to correct twin and model (RQ3); and (iv) evidence that the whole pipeline needs less real data than existing real-to-sim-to-real approaches (RQ4).

References

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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Goal. To develop a method for real-to-sim-to-real learning of **navigation models** for indoor mobile robots that reaches a given real-world navigation performance with **significantly less real data** than existing approaches. The method has three stages: (1) transfer of real-world data into a digital twin, (2) training navigation models in the twin, extended with learned world models and pretrained foundation models, and (3) transfer back to reality, where a small amount of real data corrects the twin and the model.

Thesis. Deciding at every stage which real data to acquire and where the twin can be trusted makes the whole real-to-sim-to-real pipeline for navigation at least twice as economical in real data as existing real-to-sim-to-real approaches.

How the hypotheses are decided. P is the navigation success rate in "reality". Real data (capture for the twin and real navigation trials) is counted in one operator-time cost. The hypotheses are decided



on a non-circular proxy reality built from public scans of real buildings, in which "reality" and the twin come from separate captures; a real robot validates the direction of the results. Tests, confidence bounds and scene counts are given in §9.

RQ1 (stage 1: real data → twin). Which real data is worth acquiring to build a twin for navigation?

H1. Capture guided by what matters for the navigation task and by where the twin is uncertain reaches the P of uniform capture with **at least 40% less capture**.

RQ2 (stage 2: learning in the twin). How should navigation models be trained in an imperfect twin?

H2. At the same real-data budget, training that follows the twin's uncertainty, with the twin extended by a learned world model, gives a P **at least 10 percentage points higher** than training with uniform domain randomization.

RQ3 (stage 3: twin → reality). Which few real data correct the twin and the model best?

H3. Real trials selected by the predicted twin-to-reality gap reach the target P with **at least 50% fewer trials** than randomly selected trials.

RQ4 (the whole pipeline). How much real data does the method need compared with the alternatives?

H4 (thesis). The whole method reaches the target P with **at least 2× less real data** than the strongest existing real-to-sim-to-real approach; learning from real data only is reported for reference.

A hypothesis that fails is reported as a measured curve, which still answers its question.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The dissertation is in **information and communication technology** (*informatyka techniczna i telekomunikacja*), area machine learning. Its object is the **real-to-sim-to-real pipeline for navigation models**: transferring real-world data into a digital twin, training navigation models in it and transferring them back to reality. Building simulators, world models, foundation models or benchmarks from scratch, control design and mechanical engineering are not the object of the research; the student uses existing tools, pretrained models and public data.

Key original contribution: the method. To our knowledge, the first method for real-to-sim-to-real learning of navigation models that treats the real data of the whole pipeline as **one budget spent where it matters at every stage**. It targets a given real-world navigation performance with at least two times less real data than the strongest existing real-to-sim-to-real approach (RQ4, H4). Building twins, training in simulation and using world models or foundation models are not new; deciding which real data the pipeline needs, and measuring how much it saves, is. The three stages are parts of this one method, each tested at its own step:

1. **Stage 1, real data → twin (RQ1, H1).** Capture of real data guided by what matters for the navigation task and by where the twin is still uncertain, for appearance, geometry and the physical properties needed for navigation; compared with uniform and task-blind capture.
2. **Stage 2, learning in the twin (RQ2, H2).** Training of navigation models that follows the twin's uncertainty, with the twin extended by a learned world model and the model initialized from pretrained foundation models; compared with uniform domain randomization.



3. **Stage 3, twin → reality (RQ3, H3).** Selection of the few real trials that the predicted twin-to-reality gap marks as most informative, used to correct both the twin and the model; compared with random selection.

How the method is evaluated and released (supporting the method, not contributions in their own right). Budget–performance curves of the method against existing real-to-sim-to-real approaches and learning from real data only, which give the "exchange rate" between twin and real data; a proxy-reality protocol on public scans of real buildings that counts real data exactly without a robot (a separate, higher-fidelity reference plays "reality"); and open-source code of the method, built on existing reconstruction pipelines, simulators and pretrained models.

Significance for the discipline. Real target-domain data is the main cost of deploying learned systems; methods that decide where it is worth spending, and a measured estimate of how much of it a twin can replace, are relevant to active learning, learning under distribution shift, uncertainty estimation and adaptation of pretrained models in general, beyond navigation.

Dissemination. Results are planned for peer-reviewed conferences worth 200 points and assigned to ITiT on the ministerial list of 5.01.2024: NeurIPS, ICML and ICLR (machine learning) and CVPR (computer vision), with ICCV and ECCV for resubmissions. P1 (stage 1 and first stage 2 results, RQ1–RQ2) targets NeurIPS 2027; P2 (stage 3, RQ3) ICML 2028 or CVPR 2028; P3 (the whole pipeline against existing approaches, RQ4) NeurIPS 2028, with ICLR 2029 as the fallback.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The dissertation develops **one method** with three stages: real data → digital twin → training of navigation models in the twin → transfer back to reality with a few real data that correct the twin and the model. Stages 1–3 are developed and tested one at a time (RQ1–RQ3); the whole pipeline is then tested against existing approaches (RQ4). Existing open-source tools are used throughout: no simulator, dataset or foundation model is built from scratch.

Evaluation protocol: proxy reality (decides the hypotheses). On ≥ 20 public indoor scenes of ScanNet++ [16] (10 held out from all tuning), "reality" is the laser-scan mesh textured from the DSLR images and rendered in Habitat [1] with a robot-camera model. The twin is built from a subset of the separate phone capture of the same scene, so twin and "reality" share neither images nor reconstruction method; the largest capture budget stays below the full capture. Tasks are indoor goal-reaching navigation (e.g. image-goal and point-goal; instruction-following where instructions are available), with a minimum path length set in a pilot so that P stays below its ceiling. "Reality" also has its own physical parameters (e.g. actuation noise), hidden from the method and learned only from counted real data; this physical gap is synthetic, and the robot checks it. Real data (captured views, real trials and demonstrations) is summed in one operator-time cost. The proxy's own error to held-out real images is reported, and every hypothesis is re-checked for sign with a second reference built from all DSLR images. P = success rate; path efficiency is also reported.

Stage 1 – transfer of real data into a digital twin (RQ1, H1; sem. 3–4).

- **Method:** the twin's appearance and geometry come from neural scene reconstruction (3DGS-type [10]), run as a mesh in the simulator (as in [11]) or rendered directly; its physical part (e.g. traversable surfaces, collisions, the robot's actuation noise) is estimated from a few measurements as a posterior [15]. The next views are chosen where the twin's uncertainty (e.g. Fisher information [17] or an uncertainty field [18] adapted to 3DGS) is highest in the regions that matter for navigation: along planner paths between likely goals and, optionally, where a pretrained vision-language model grounds the task. Candidates come from the recorded capture; capture time is estimated as a walking tour.



- *Comparators*: uniform capture; task-blind and risk-weighted uncertainty selection [17, 19]; vision-language-guided reconstruction-quality selection [20].
- *Criterion (H1)*: budget–P curves (≥ 4 budgets, the same navigation learner trained in each twin); $\geq 40\%$ less capture (in views; operator time also reported) than uniform at equal P, with the 95% bound of the ratio below 1.

Stage 2 – training navigation models in the twin (RQ2, H2; sem. 4).

- *Method*: the navigation model is initialized from a pretrained foundation model (e.g. a general navigation model or a vision-language-action model [21–24]) and trained in the twin by imitation of planner demonstrations and then by reinforcement learning on a subset of scenes, with parameter-efficient adaptation where the model is large. Randomization strength follows the twin's uncertainty per region and per physical parameter, and twin samples far from a few real samples in a frozen-encoder space [29] are down-weighted; these real samples are held out from the capture and count towards its budget. A learned world model (e.g. a video world model [25, 26]) adapted on twin data generates additional trajectories where the twin is most uncertain.
- *Comparators*: uniform randomization of appearance [5] and dynamics [6], and none; ablations without the world model, without uncertainty weighting and without foundation-model initialization.
- *Criterion (H2)*: mean paired gain in P over uniform randomization ≥ 10 pp on held-out scenes, lower 95% bound > 0 , at each of ≥ 2 capture budgets.

Stage 3 – transfer back to reality with few real data (RQ3, H3; end of sem. 4 – sem. 5).

- *Method*: candidate real trials are scored by the predicted twin-to-reality gap, combining the twin's uncertainty along the planned path, disagreement between twin and world model, and the navigation model's own uncertainty (ensemble spread). The top-ranked trials are run in reality; they correct the twin (re-capture where real and twin observations disagree, counted in the budget; posterior update of physical parameters [7, 15]) and the model (co-training with twin data [9]). Real P from few trials is estimated with prediction-powered inference [30].
- *Comparators*: random selection, uniform coverage and failure-prone selection from the twin alone [27].
- *Criterion (H3)*: the target P with $\geq 50\%$ fewer trials than random selection (95% bound of the ratio below 1).

Whole pipeline (RQ4, H4; sem. 6). Budget curves (real data against P) of the whole method, of the strongest existing real-to-sim-to-real approach, re-implemented for navigation and tuned with the same effort as the method (uniform capture, uniform randomization, training in the twin and random or failure-driven real trials, with the same reconstruction pipeline, learner and correction; e.g. [14, 27]), and of learning from real data only (imitation of real demonstrations, same initialization). The method's settings are fixed before this stage. *Criterion (H4)*: $\geq 2\times$ less real data than the existing approach at the target P, with the 95% bound of the ratio below 1, and the stage 1–3 effects keep their sign.

Real-robot validation. Two campaigns (sem. 5 and 7) with own phone or RGB-D captures of 2 rooms at PWR and a mobile robot: uniform capture versus the method, ~ 8 robot-hours per campaign; they check the direction of the results and never tune thresholds.

Statistics and compute. Targets are defined from each comparator's own curve at its largest budget. Tests are one-sided at $\alpha = 0.05$ with a scene-level bootstrap and Holm correction, pre-registered before each stage. Training runs on WCSS and PLGrid GPUs (computing grant applied for in sem. 3); smaller models are used for development. Code is released where dataset and model licences permit.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać



cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Roboty, które poruszają się po budynkach, na przykład w magazynach, szpitalach czy biurach, coraz częściej korzystają z modeli nawigacji, które uczą się na podstawie doświadczenia. Aby taki model dobrze działał w konkretnym budynku, potrzebuje wielu danych z tego miejsca, a zbieranie ich w rzeczywistości jest powolne i kosztowne. Dlatego coraz częściej robot uczy się w cyfrowym bliźniaku: wiernej kopii prawdziwego budynku, którą można odtworzyć z krótkiego nagrania i kilku pomiarów. Powstaje w ten sposób droga w trzech krokach: zbieramy dane w rzeczywistości i budujemy z nich kopię, uczymy w niej model nawigacji, a następnie przenosimy go z powrotem do rzeczywistości i tam sprawdzamy oraz poprawiamy. Każdy z tych kroków wymaga jednak prawdziwych danych.

Celem rozprawy jest opracowanie metody, która przejdzie tę drogę z dużo mniejszą ilością prawdziwych danych niż dotychczasowe podejścia. W pierwszym kroku, zamiast nagrywać wszystko po równo, będziemy nagrywać przede wszystkim te miejsca, które są ważne dla poruszania się robota, a o których kopia wie jeszcze za mało. W drugim kroku, ponieważ kopia nigdy nie jest idealna, będziemy uczyć model tak, aby nie polegał na jej niepewnych fragmentach; pomogą w tym model świata, czyli program, który uczy się przewidywać, co wydarzy się dalej, oraz duże modele wstępnie wytrenowane na wielu innych danych. W trzecim kroku wybierzemy tylko te nieliczne próby w rzeczywistości, które najwięcej mówią o tym, gdzie kopia i model jeszcze się mylą, i wykorzystamy je do ich poprawienia. Badania będą prowadzone przede wszystkim na publicznie dostępnych skanach prawdziwych budynków, a wyniki zostaną potwierdzone na prawdziwym robocie.

Spodziewanym efektem jest metoda, która pozwoli nauczyć robota poruszania się w nowym budynku przy dużo mniejszej ilości prawdziwych danych, oraz pomiar tego, ile prawdziwych danych pozwala ona zaoszczędzić w porównaniu z dotychczasowymi podejściami. Może to przyspieszyć i potanić wdrażanie robotów mobilnych w magazynach, szpitalach i biurach. Opracowane oprogramowanie zostanie udostępnione publicznie.

Abstract for general public

Robots that move around buildings, for example in warehouses, hospitals or offices, increasingly rely on navigation models that learn from experience. For such a model to work well in a particular building, it needs a lot of data from that place, and collecting it in reality is slow and costly. That is why robots increasingly learn in a digital twin: a faithful copy of a real building that can be rebuilt from a short recording and a few measurements. This creates a path in three steps: we collect data in reality and build the copy from it, train the navigation model in the copy, and then transfer it back to reality, where we check and correct it. Each of these steps, however, needs real data.

The goal of this dissertation is to develop a method that follows this path with much less real data than existing approaches. In the first step, instead of recording everything evenly, we will record mainly the places that matter for the robot's movement and about which the copy still knows too little. In the second step, because the copy is never perfect, we will train the model so that it does not rely on the uncertain parts of the copy; a world model, a program that learns to predict what happens next, and large models pretrained on much other data will help with this. In the third step, we will choose only the few trials in reality that tell the most about where the copy and the model are still wrong, and use them to correct both. The research will be carried out mainly on publicly available scans of real buildings, and the results will be confirmed on a real robot.

The expected result is a method that makes it possible to teach a robot to move around a new building with much less real data, and a measurement of how much real data it saves compared with



existing approaches. This may make deploying mobile robots in warehouses, hospitals and offices faster and cheaper. The software developed will be made publicly available.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027 / maj 2027

Paper **P1** submitted for review to the main track of the **Conference on Neural Information Processing Systems (NeurIPS 2027)**. NeurIPS is peer-reviewed international conference proceedings worth **200 points** on the ministerial list (conference Lp. 87), assigned to the discipline *informatyka techniczna i telekomunikacja*. P1 reports the first two stages of the method for navigation models (stage 1 and the first stage 2 results, RQ1–RQ2): capture of real data guided by the navigation task and by the twin's uncertainty (H1), and training of navigation models that follows the twin's uncertainty (first H2 results). Both are evaluated on public scans of real buildings by the real data they save (task T4.2). If it is rejected, the revised paper goes to **CVPR 2028** (expected deadline November 2027, also 200 points), and then to the next 200-point conference (§12).

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

Planned outputs by year (200-point conferences assigned to the discipline, list of 5.01.2024).

Year	Papers (planned submission, 200 points each)	Code	Grants	Mobility, events
1 (2025/26)	—	—	—	—
2 (2026/27)	P1 (stage 1 and first stage 2 results, RQ1–RQ2): NeurIPS 2027, May 2027 (§11)	Pipeline on public scans	SzD Minigrant; NCN PRELUDIUM; NAWA Bekker	Summer school poster



3 (2027/28)	P2 (stage 3, few real data, RQ3): ICML or CVPR 2028; P3 (the whole pipeline, RQ4): NeurIPS 2028	Stage 3 code	—	Mid-term; 3-month visit (sem. 6)
4 (2028/29)	Optional P4, consolidated study: ICLR 2029 or CVPR 2029	Method release	—	Dissertation

Foreign research visit (candidate hosts, not yet contacted). (1) Z. Kira's group, Georgia Tech: EmbodiedSplat [11]. (2) Multi-robot Systems Lab, Stanford University (M. Schwager): Gaussian-splatting scene models. A European host (Erasmus+ route) working on active reconstruction or sim-to-real transfer, and a foreign co-author for P2, are sought with the supervisor in sem. 4.

Collaboration at PWr. Robot validation is planned with the K29 *Denali* laboratory (Pioneer 3-DX, Jaguar 4x4, ROS 2); a written agreement is task T3.2 (Nov 2026).

Risks and mitigation (semester affected).

- *One stage gives no gain* (3–5): its budget curve still answers its RQ; H4 is still tested.
- *Twin uncertainty is poorly calibrated* (3–4): two estimators and an ensemble fallback.
- *Proxy reality is easier than reality* (4–7): separate captures for twin and reference, a second reference and robot validation check the direction (§9).
- *Learning from real data only is strong at small budgets* (6): in navigation, gains from more data in a known location saturate quickly [8]; H4 is reported as a full budget curve.
- *Compute* (3–7): WCSS and PLGrid grants in sem. 3; smaller models for development; fewer seeds.
- *Crowded field, scooping* (3–6): twins, world models and fine-tuning in simulation are active areas, with targeted or real rollouts [27, 28]; we claim only the real-data budget; early preprints.
- *Licences, robot access* (3, 5, 7): gated or restricted models have open alternatives; without a robot, validation moves by a semester (the proxy reality decides).
- *Rejection at a top venue* (4–7): each paper has a resubmission path inside the 200-point set 2–4 months later (NeurIPS → CVPR → ICML → ECCV/ICCV).

Future applicability. The method is not specific to navigation; robotic manipulation is a natural extension after the dissertation.

Prior work (background only). A pre-PhD NLP paper (ACL 2025 SRW, forecasting from hidden states); not part of the dissertation.

Ethics and data. Own captures are anonymized (GDPR); public datasets are used under their licences.

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor

Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

- **Weekly meeting** (~1 h, in person at K46 or online via MS Teams) to discuss progress, results and next steps.
- **Monthly written progress summary** (completed tasks, results, problems, plan). It lists every IPB task from §3 with its **% completion**, in the same form as the task table of the mid-term self-report.
- **Semester review** against the IPB schedule (§3), before each annual report and the mid-term evaluation. Deviations from the plan and their reasons are recorded and agreed with the supervisor.
- **Supervisor co-signs** the annual reports and the mid-term self-report (autoreferat).



- **Paper-writing sprints** before each submission deadline, with joint editing in Overleaf. The supervisor co-authors and reviews the papers before submission.
- **Ongoing communication** via e-mail / Teams chat. The code and experiment logs are in a shared Git repository that the supervisor can access.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor